

How the bus works, and what it returns

One workflow per unit. Agents write evidence. The orchestrator derives the status and the final outcome. The 8 October stress run is a crash test of that bus, not a five-agent vision score.

200 workflows

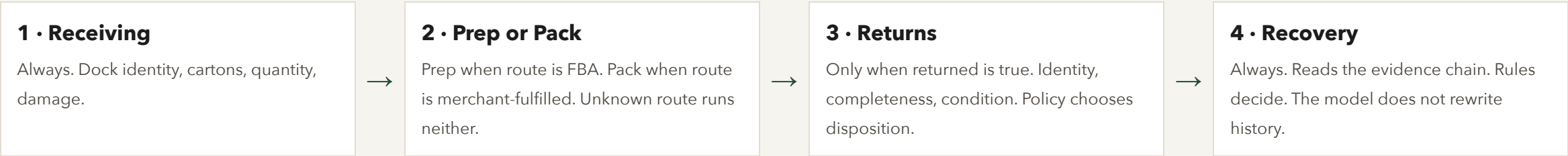
0 crashes

0 schema failures

No Claude calls

Receiving always. Prep or Pack by route. Returns only if returned.

flow.json, flow_id standard-v1. A stage that does not apply is skipped and the reason is stored. A stage is never skipped because an agent was down. That is an error.



POLICY	DEFAULT	WHAT IT DOES
timeout	30 seconds	Then a retry.
retries	1	Transient failures retry. A 4xx refusal does not.
on_uncertain	continue	UNCERTAIN stays UNCERTAIN and travels forward. A person is asked when needs_human is true.
on_error	continue	Other stages can still write evidence. The workflow still ends FAILED if a required stage never finished.

Every agent output is checked for schema, stage, workflow, tenant, and hash before it is stored. Evidence is append-only. A person’s disagreement is an override that names the record it supersedes.

A workflow, an evidence chain, and a derived outcome

The latest agent answer is not the source of truth. rollup.py derives status and final outcome from the stored records and overrides.

WF-...

workflow_id from org and unit. Starting the same unit again advances it. It does not duplicate it.

Evidence

One immutable record per attempt. Failed attempts stay. Retries add a new record.

Outcome

CLEAN, EXCEPTION, NEEDS_REVIEW, CLAIM_RECOMMENDED, or INCOMPLETE, with a reason and the record ids behind it.

FIELD ON THE FINISHED WORKFLOW	WHAT YOU SHOW
status	PENDING, IN_PROGRESS, COMPLETED, FAILED, BLOCKED, or RECOVERY_REQUIRED
stage_results	Per stage: skipped, completed, pending, or error, plus verdict, attempts, and error code
final_outcome.outcome	The business result
final_outcome.reason	A sentence naming the stages that drove it
effective_verdicts	Each stage after overrides
claimable_usd	Set only when the outcome is CLAIM_RECOMMENDED
provisional	True whenever status is not COMPLETED
needs_human	True when a person was asked, or a required stage did not finish

Status says whether the bus finished. Outcome says what it found.

PASS
Record it. Continue.

FAIL
A finished judgment, not a crash. The bus keeps going.

UNCERTAIN
Kept as written. Never turned into PASS.

IF THE EVIDENCE SAYS	STATUS	FINAL OUTCOME
Every required stage passed	COMPLETED	CLEAN
A stage FAIL, and Recovery did not contradict a charge	COMPLETED	EXCEPTION
A stage asks for a person	BLOCKED	NEEDS_REVIEW
FAIL and a person is also waiting	BLOCKED	EXCEPTION. The person comes first in the status.
Recovery verdict FAIL, earlier stages finished	COMPLETED	CLAIM_RECOMMENDED. claimable_usd is set.
Recovery silent, needs_human false	COMPLETED if the rest passed	CLEAN. Silence is not a failure.
A required stage timed out, crashed, or refused	FAILED	INCOMPLETE. Resume retries the failed stage.

A claim is withheld when an earlier stage did not finish, even if Recovery’s own verdict was FAIL. That outcome becomes INCOMPLETE.

60 completed, 10 blocked, 30 failed. The process stayed up.

These units are scripted inside the orchestrator. No Claude. No ports 8102-8105. Route rotates through FBA, merchant-fulfilled, and unknown. Half are marked returned. Ten units in each group.

SCENARIO	STATUS	OUTCOME	WHAT IT PROVES
all_pass	COMPLETED 10	CLEAN 10	A full pass finishes clean.
receiving_fail	COMPLETED 10	EXCEPTION 10	A FAIL is a judgment. The bus continues.
midstage_fail	COMPLETED 10	CLEAN 4, EXCEPTION 6	Unknown route has no Prep or Pack, so nothing mid-stage to fail.
uncertain_needs_person	BLOCKED 10	NEEDS_REVIEW 10	The bus continues, then the rollup blocks because a person was asked.
recovery_claim	COMPLETED 10	CLAIM_RECOMMENDED 10	A claim is allowed when earlier stages passed.
recovery_silent	COMPLETED 10	CLEAN 10	Silent uncertainty with needs_human false stays CLEAN.
timeout_then_pass	COMPLETED 10	CLEAN 10	Receiving tried twice (20 attempts). The retry passed.
timeout_exhausted	FAILED 10	INCOMPLETE 10	Both attempts used. Stage stays pending. Process stays up.
agent_crash	FAILED 10	INCOMPLETE 10	RuntimeError inside Recovery is stored as UNCERTAIN. It does not escape.
schema_refusal	FAILED 10	INCOMPLETE 10	A bare HTTP 422 stops the bus. Later stages are not called.

73 completed, 27 blocked. Dummy rows, replayed.

Source: data/sample. The sample README marks it dummy. A stage runs only when the case is eligible. Receiving and Recovery always. Prep for FBA. Pack for merchant-fulfilled. Returns only when returned is true.

48
CLEAN

31
EXCEPTION

21
NEEDS_REVIEW

0
Claims from these rows

ELIGIBILITY	UNITS	REPLAY RULE
Neither Prep nor Pack	9	Unknown route.
Prep ran	62	FAIL on missing, unsealed, obscured, or badly placed marks.
Pack ran	29	PASS on seal. FAIL on stop_and_fix.
Returns ran	24	PASS on a disposition. pending_review is UNCERTAIN.
Receiving	100	PASS only when identity is yes and both damage fields are none.

Six units are BLOCKED and EXCEPTION together: one stage FAIL, another asked for a person. Twenty-one are BLOCKED / NEEDS_REVIEW with no FAIL. Twenty-five are EXCEPTION with nobody waiting. Recovery does not recommend a claim from these CSVs. A weight, lost, or warehouse line is silent. An inbound-defect or refund line asks for a person, because the CSV has no photo.

All ten blocked. Pack has no row to replay.

These dummy units have a Receiving row, a Prep row, a Returns row, and a fee row. This pass uses flow.pod4-tracers.json, so Pack is called anyway and returns UNCERTAIN. Not a Claude run.

UNIT	PRODUCT	OUTCOME	RECEIVING	PREP	PACK	RETURNS	RECOVERY
0003	Jigsaw Puzzle	EXCEPTION	UNCERTAIN	FAIL	UNCERTAIN	PASS	UNCERTAIN
0014	LED Desk Lamp	NEEDS_REVIEW	PASS	PASS	UNCERTAIN	PASS	UNCERTAIN
0018	Steel Water Bottle	NEEDS_REVIEW	PASS	PASS	UNCERTAIN	PASS	UNCERTAIN
0030	Vitamin C Serum	EXCEPTION	PASS	FAIL	UNCERTAIN	PASS	UNCERTAIN
0036	Jigsaw Puzzle	EXCEPTION	FAIL	PASS	UNCERTAIN	PASS	UNCERTAIN
0038	Whey Protein	NEEDS_REVIEW	PASS	UNCERTAIN	UNCERTAIN	PASS	UNCERTAIN
0039	Vitamin C Serum	NEEDS_REVIEW	PASS	PASS	UNCERTAIN	PASS	UNCERTAIN
0041	Nylon Dog Leash	NEEDS_REVIEW	PASS	PASS	UNCERTAIN	PASS	UNCERTAIN
0048	Soy Candle Trio	NEEDS_REVIEW	PASS	PASS	UNCERTAIN	PASS	UNCERTAIN
0050	Jigsaw Puzzle	EXCEPTION	FAIL	UNCERTAIN	UNCERTAIN	PASS	UNCERTAIN

Workflow status is BLOCKED on all ten. Returns replayed PASS on all ten. Recovery stayed UNCERTAIN: either a silent fee line, or a charge with no photo that can support or contradict it.

The judgment that already landed is kept. The process does not die.

BREAK	WHAT IS STORED
Connection refused	FAILED / INCOMPLETE. Receiving UNCERTAIN, error agent_unavailable. Prep is not asked to invent that judgment.
Resume after the agent answers	On the injected case: COMPLETED / CLEAN. Receiving PASS. Attempts seen: 5.
Every agent raises	FAILED / INCOMPLETE. Four stages error / UNCERTAIN. The exception does not leave run_workflow.
Prep returns a bare 422	FAILED / INCOMPLETE. Receiving PASS is kept. Pack, Returns, and Recovery are not called.
UNIT-0014, Receiving down, then resume	First pass FAILED / INCOMPLETE. After resume: BLOCKED / NEEDS_REVIEW, because Pack and Recovery are still UNCERTAIN on that lamp.

2

Receiving attempts while the agent is down, then stop.

Kept

A finished PASS stays on the evidence chain through a later refusal.

Resume

Retries errored stages on the same workflow id.

The bus was measured. A five-stage Claude score was not.

200

Workflows returned. 0 crashes. 0 schema failures.

10

Scripted claims recommended, only when earlier stages had passed.

0

Claims from the dummy CSV. No photo, no claimable dollars.

SAY THIS	LEAVE THIS OFF
The orchestrator owns the outcome. Agents contribute evidence.	A single Pod 4 accuracy percentage.
FAIL continues. A crash, timeout, or 422 is stored and can be resumed.	Calling the stress run a vision evaluation.
CLEAN, EXCEPTION, NEEDS_REVIEW, CLAIM_RECOMMENDED, and INCOMPLETE are the five results a workflow can give.	UNIT-001, UNIT-002, and UNIT-003. They were not in these 200.
Stage grades live on the other deck: identity, prep fields, false seals.	A false-claim rate. There is no labeled fee sheet large enough to score one.

Sources: docs/pod4-stress-run.md, docs/evaluation.md, docs/pod4-ten-eligible.md, orchestration/flow.json, orchestration/rollup.py. Tracer photo folders are empty.